Introduction & Problem Statement
HireHub

Internship Discovery and Recruitment Management

Problem

Separate forms, emails and spreadsheets make candidate review and application tracking difficult to coordinate.

Our solution

A shared portal for students, recruiters and administrators, covering discovery, applications, interviews and communication.

React + TypeScript FastAPI + SQLAlchemy MySQL

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Functionalities & Associated DBMS Concepts

Functionality	DBMS concepts used in HireHub
Registration & profiles	Primary keys, unique emails, foreign keys and one-to-one profiles
Search & recommendations	Filtered queries, pattern matching, sorting and pagination
Applications & bookmarks	Linking tables for student–internship relationships
Student skills	Many-to-many relationship through `student_skills`
Candidate review & interviews	Related inserts and updates, status fields and commits
Messages & notifications	Sender/receiver foreign keys, timestamps and read flags
Admin analytics	`JOIN`, `COUNT`, `AVG` and `GROUP BY`

Database: 12 tables connected by 16 foreign-key relationships.

Modules & Explanations

Module	Responsibility
Student	Academic profiles, skills, document uploads, internship discovery, bookmarks and application tracking
Recruiter	Internship posting and management, candidate profiles, resume access, table/Kanban applicant views and status updates
Interviews	Date, duration, interview type and meeting link; scheduling updates the application and creates a notification
Communication	Student–recruiter conversations connected through applications, message previews and unread counts
Administrator	User and posting management, application oversight, statistics, charts and recent activity

Results & Discussion

Implemented outputs

- **Student dashboard:** application progress, saved roles and recommended internships.
- **Recruiter workspace:** applicant pipeline, candidate details, resume access and interview list.
- **Admin dashboard:** 14-day application trends, status distribution, work modes and company summaries.
- **Communication:** stored conversations and notifications for recruitment events.

Discussion

The same application record connects student tracking, recruiter review and database reports.

Evaluation scope: demonstration data is available. Large-scale performance and placement impact have not been measured.

Advantages & Limitations

Advantages

- One system for applications, interviews and communication.
- Visible application stages for both sides.
- Foreign keys connect recruitment records.
- Reports use the stored recruitment data.

Limitations

- Recommendations use simple matching rules.
- External applications are not automatically tracked.
- Base64 resumes increase database storage.
- Calendar grid and some settings remain incomplete.

Next improvements: stronger database constraints, normalized internship skills and testing with more users.

Project Highlights & Expected Impact

Highlights: linked application and interview records, application-connected messaging, rule-based suggestions and admin analytics.

Manual workflow baseline	HireHub implementation
Candidate details across forms and emails	Combined candidate and application view
Status maintained in separate spreadsheets	Shared application-status record
Interview details coordinated separately	Interviews linked to applications
Reports prepared by manual counting	Aggregate database queries

Contribution: integration of recruitment workflows within one project.

Expected impact: less repeated coordination and clearer application visibility.

Manual-process comparison only. A user study is needed to measure impact.

Challenges & Learnings

Challenge	Learning
Modeling recruitment relationships	Choose one-to-one, one-to-many and many-to-many relationships from the actual workflow
MySQL foreign-key setup	Related columns need compatible types and collations
Uploading large resumes	Use multipart requests and suitable database column types
Managing user roles and ownership	Check permissions in the backend as well as the interface
Coordinating related updates	Plan transaction boundaries and enforce important rules in the database

Key learning: a good data model must work with reliable API and interface behavior.

Budgeting

Local academic demonstration: estimated additional cash cost

Component	Assumption	Cost
Development tools	Existing environment and open-source tooling	INR 0
MySQL database	Local community installation	INR 0
Hosting and domain	Existing laptop; no paid hosting or domain	INR 0
Hardware / internet	Existing resources; electricity excluded	Excluded
Development effort	Student project contribution	Excluded
Additional cash total	**Under the assumptions above**	**INR 0**

Future deployment: budget separately for hosting, database storage, backups and an optional domain.

Planning estimate only. Actual expenses have not been supplied.